Changes to Allele Frequencies in Gene Pools



Introduction

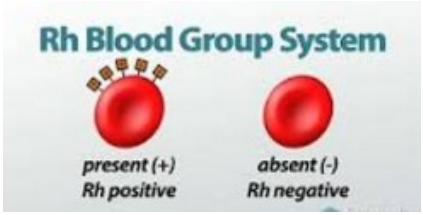
- Allele frequency can change within gene pools
- This can be due to:
  - Gene flow between populations
  - Random genetic drift including
    - Founder Effect
    - Bottleneck Effect
  - Natural Selection

Learning Aim: List reasons why allele frequency may change within gene pools.

Gene Flow and Allele frequency

- Flow of genes from one population to another with human migration and inter-breeding.

  - Example: Rhesus factor in Chinese population:
    - In the past, Chinese population was Rh negative (Rh-): no rhesus factor on blood cells.
    - European traders and sailors in 16th century carried allele for Rhesus factor (Rh +), and interbred with Chinese population.
    - Some offspring then were Rh+



    - Frequency of Rh + allele is still low, but greater than it was pre-European gene flow.

Learning Aim: Define and give examples of gene flow.

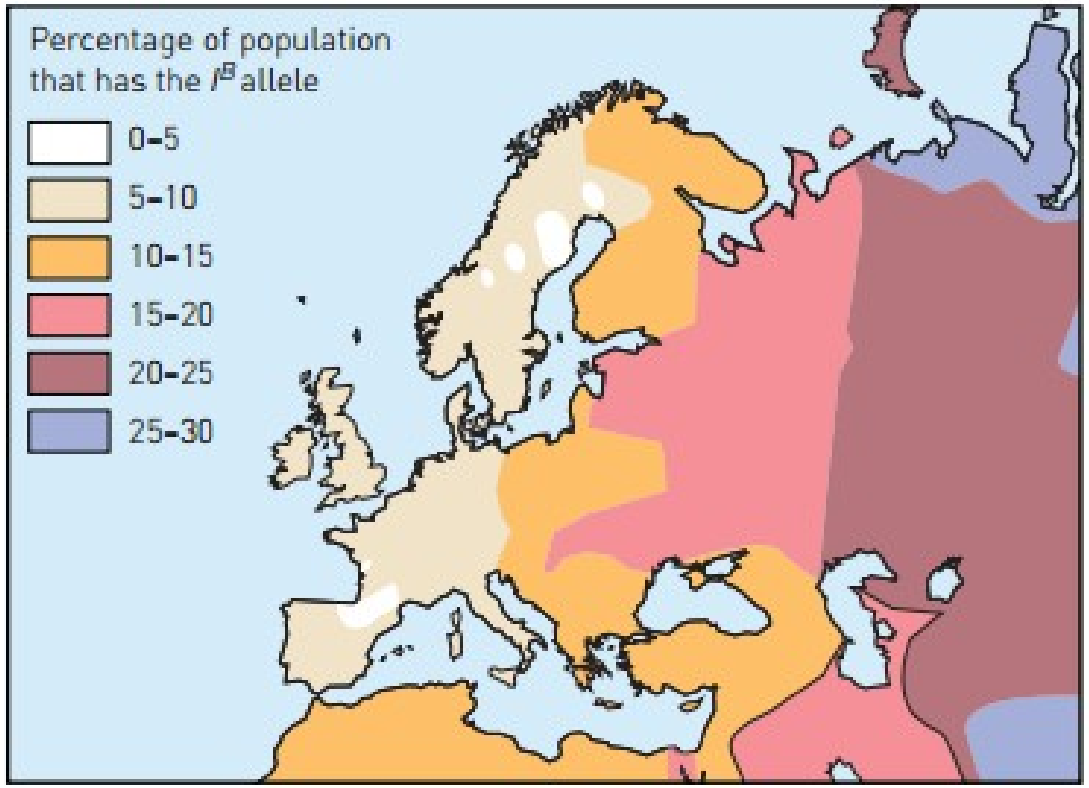
Gene Flow and Allele frequency

Example: ABO blood group distribution:

- East Asian populations have greater frequency of B group allele.

- Spread into Europe during Mongol invasions of 12th and 13th Centuries.

- B group more common in Eastern Europe, decreasing in frequency moving west, in line with exposure to Mongol invaders.



Learning Aim: Define and give examples of gene flow.

Barriers to Gene Flow

- Things that prevent contact and/or interbreeding between populations:
- Geographical Barriers
  - mountain ranges, oceans, big lakes, deserts, ice sheets
  - Isolation over time means different gene pools develop
  - Natural selection (more on this later) means advantageous traits for the environment become more common
- Socio-cultural Barriers
  - Different religion, language, cultural practices, social position can also prevent interbreeding between populations
  - Historical practices of cousin marriage in the Mediterranean to preserve property and power (cultural isolation) have led to a greater frequency of the allele for Beta Thalassemia.
  - Eg Basque people of the Pyrenees mountains: geographical and cultural isolation → distinctive appearance, unusual blood group allele frequencies.



Learning Aim: Discuss Barriers to gene flow and provide examples.

Barriers to Gene Flow - Disadvantages and Advantages

Disadvantages

- Greater incidence of genetic disease
  - Small/isolated populations:
  - Marriage between close relatives, eg cousins, are common
  - Means two carriers of a genetic disease are more likely to marry
  - Therefore greater incidence of the genetic disease

- Advantageous genes won't enter population through migration

Advantage

- Barriers to gene flow mean that lethal recessive alleles from other populations are less likely to enter the isolated population.

Learning Aim: List disadvantages and advantages of barriers to gene flow.

Genetic Drift

- Allele frequency change by ***chance***, rather than by natural selection.
- Random nature of reproduction can mean some alleles become more frequent by chance.
- Happens in all populations, but can have a larger effect in small populations.



Figure 14.3 Simplified representation of genetic drift in a very small population. If the two red-haired individuals in the first generation mated and had four children, and the other two individuals mated and had only two children, the proportion of red-haired individuals in the population would increase.

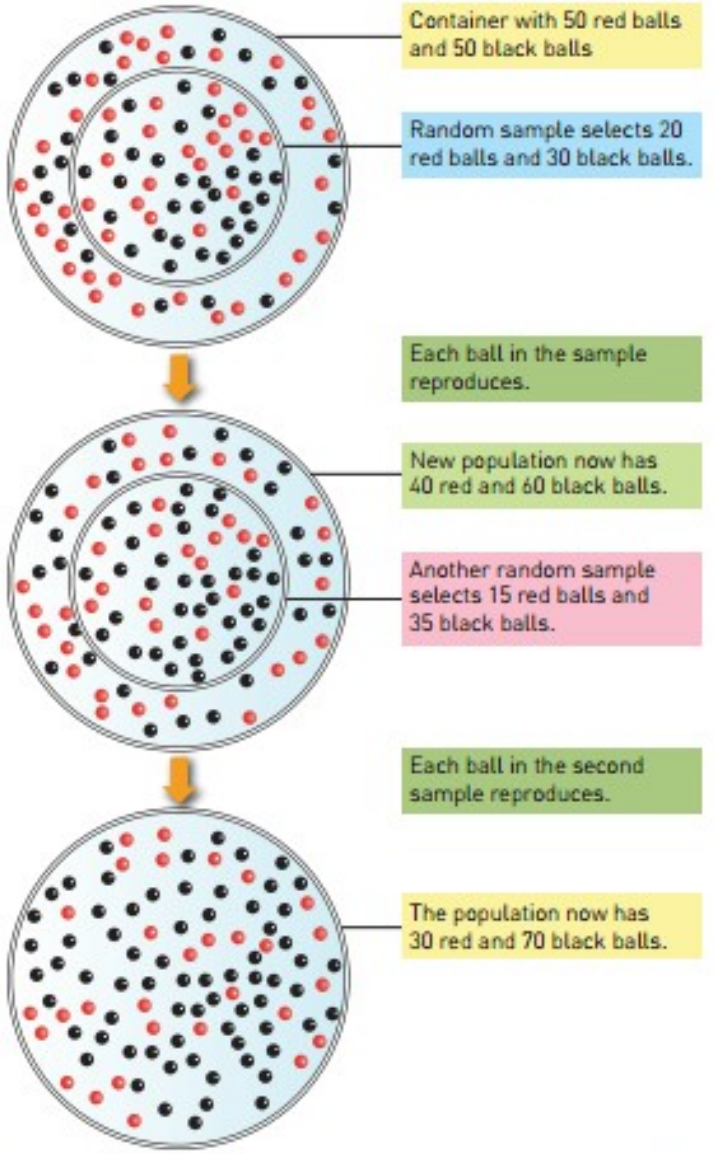


Figure 14.2 A model showing how random sampling of a pair of alleles can, by chance, increase the frequency of one allele and decrease the frequency of the other

Learning Aim: Define and briefly explain genetic drift.

Genetic Drift is Random

- Allele frequency changes due to random/chance factors eg:
  - A subset of the population becoming isolated
  - Migration and interbreeding between populations
  - More people with a particular allele having more children through random chance
- Change of frequency is neither an advantage nor a disadvantage

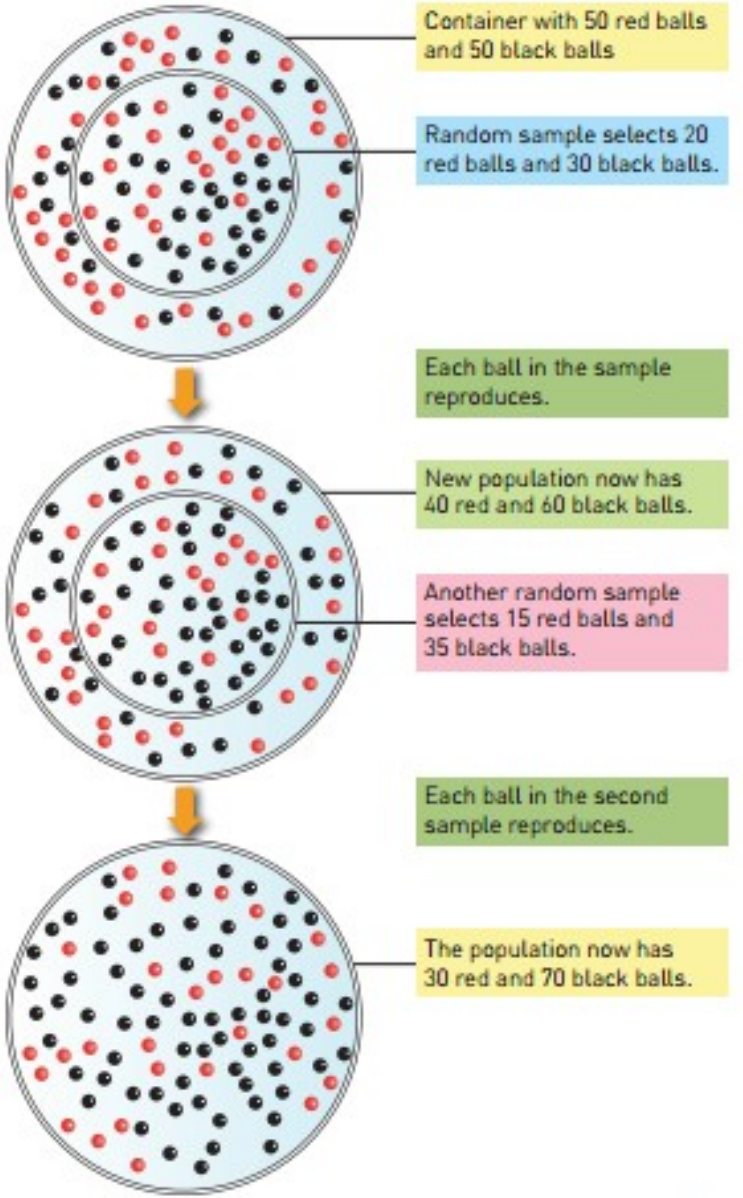


Figure 14.2 A model showing how random sampling of a pair of alleles can, by chance, increase the frequency of one allele and decrease the frequency of the other

Learning Aim: Describe how genetic drift is random.

Genetic Drift is Non-Directional

- Frequency of a particular allele can increase and decrease over time due to random factors :

  - Eg: More red-headed people might randomly have kids in one generation, increasing the allele frequency for red hair, then the next generation they may have less, decreasing the frequency. The generation after that, there may be more again.



Figure 14.3 Simplified representation of genetic drift in a very small population. If the two red-haired individuals in the first generation mated and had four children, and the other two individuals mated and had only two children, the proportion of red-haired individuals in the population would increase.

Learning Aim: Describe how genetic drift is non-directional.

Genetic Drift is Non-Adaptive and not influenced by the environment

- The change in allele frequency doesn't cause an adaptive advantage in the environment
  - Doesn't increase chance of survival or reproduction.

  - Eg: an random increase in the allele frequency for straight hair doesn't cause an advantage. The next generation there may be a random decrease based on the randomness in reproduction.

Learning Aim: Describe how genetic drift is non-adaptive and not influenced by the environment.

Evidence for Genetic Drift – The “Dunkers”

- USA religious sect in Pennsylvania – similar to Amish
  - Originally from a population of German Baptists from Hesse in Germany – arrived in USA approx 300 years ago.
  - Only marry and reproduce within their community.
- Studies done to comparing allele frequencies Dunkers with surrounding American community AND modern day Germans from Hesse
  - Dunkers have different allele frequencies to modern day Germans from Hesse AND surrounding American community.
  - Environment of surrounding mainstream USA community the same as for Dunkers, therefore differences in allele frequency must be due to random drift as natural selection can’t be the reason – same environment.



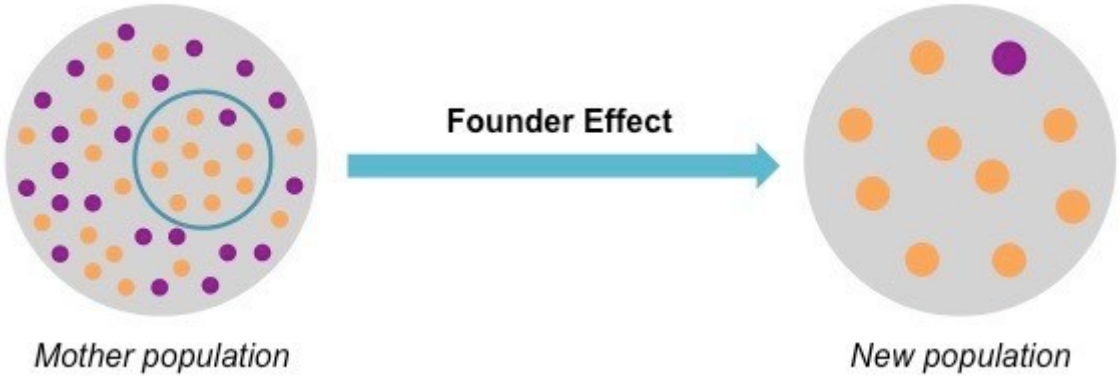
Learning Aim: Give an example of evidence for genetic drift.

Genetic Drift Example: The Founder Effect

- Occurs when a small group moves away from homeland and establishes an isolated community, which later breeds and expands

  - Small sample of homeland population is not likely to have exact same allele frequency as whole population.

  - As isolated population expands, so does the new allele frequency.



Learning Aim: Define and explain the Founder Effect

Genetic Drift "Founder Effect" example: Pitcairn Islanders

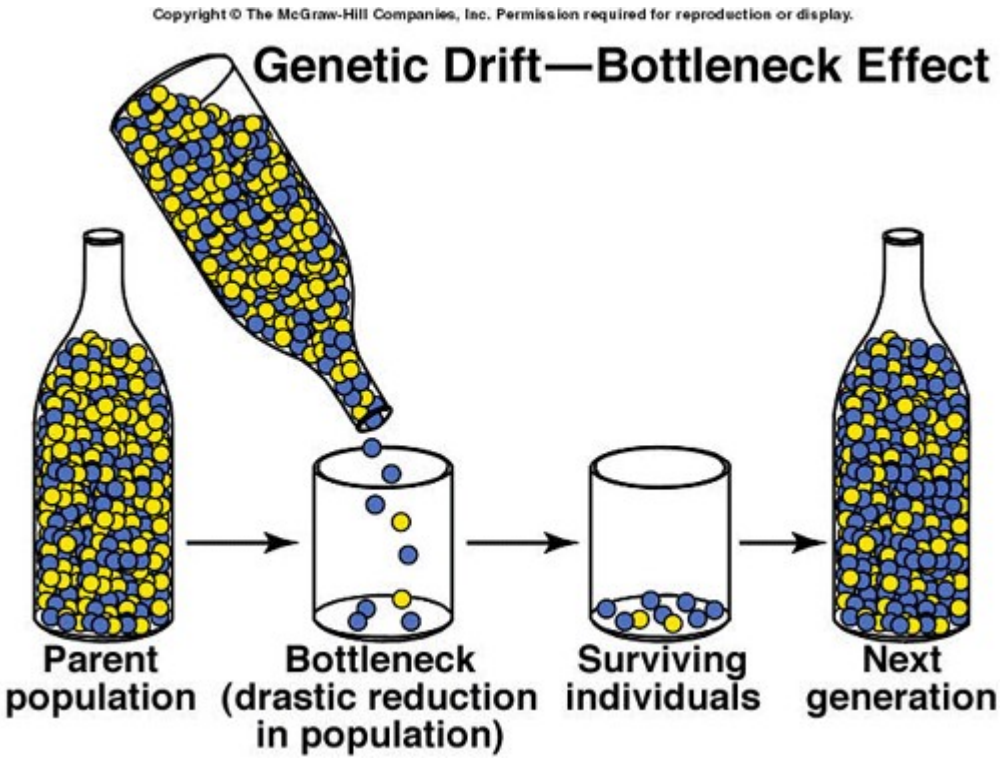
- In 1790, Pitcairn Island was populated by:
  - 9 mutineers from the Bounty
  - 6 men, 12 women Polynesians
- Geographical isolation, few newcomers
  - Descendants show less genetic diversity (and fewer surnames), and different gene frequencies compared to general populations.
- Pitcairn Island descendants at greater risk of cardiovascular disease due to this – the original inhabitants would have randomly had a greater allele frequency for cardiovascular disease.



Learning Aim: Give an example of the Founder Effect

Genetic Drift Example: The Bottleneck Effect

- Occurs when natural disaster severely reduces population size of an isolated population:
  - Small sample of homeland population is not likely to have exact same allele frequency as whole population.
  - As isolated population expands, so does the new allele frequency.



Learning Aim: Define and explain the Bottleneck Effect

Bottleneck effect example: Pingelap

- Pingelap: island in Micronesia
- 1775: typhoon wiped population down to 20 individuals
- Among survivors: one person who carried allele for *achromatopsia* (total colour blindness)
- After multiple generations:
  - incidence of achromatopsia on Pingelap is 5%
  - 30% of population carriers.
  - Compare to rest of world: incidence 0.0033%





Learning Aim: Give an example of the Bottleneck Effect.